# Nishal Stanislaus Silva, *PhD*

Machine Learning Research Engineer | Real-Time Inference, Signal Processing, Embedded C++ | Applied ML

*Canadian Permanent Resident, Eligible to work in Canada without sponsorship*

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18th October 2026

**Re: Senior / Staff Software Engineer, High-Performance Onboard Algorithms – Waabi**

Dear Hiring Manager,

I am applying for the Senior / Staff Software Engineer, High-Performance Onboard Algorithms role at Waabi. My background is real-time, low-latency software on constrained hardware, which is the core of what this team builds for autonomous trucks. I hold a PhD (University of Trento, 2025) and have 10+ years across industry and academia, and I am drawn to a role where sensor data has to be processed fast and reliably on target hardware.

Three parts of your posting match my work directly. For real-time performance on target hardware, my published system runs at 14 ms on a Raspberry Pi 4 at about 31% CPU, 74x faster than the DTW baseline at F1 0.76 (Journal of the Audio Engineering Society, 2026). I profiled CPU use against a strict latency budget and benchmarked accuracy, latency and CPU cost through baselines and ablations. For compute-intensive native code, I wrote Nebula, an open-source C++17 feature-extraction library with per-feature latency benchmarking on ARM, and wrote C++ and Python real-time inference code for embedded and IoT hardware at MAS Holdings. For image and sensor pipelines, I built OpenCV-based inspection systems that integrate camera and sensor streams into production, including a care-label checker that cut inspection time by 99.5% and raised efficiency by up to 300%.

I should be direct about the gaps. I have not shipped Rust, CUDA, Vulkan or OpenCL code, and I have not worked with LiDAR or radar, or written SIMD intrinsics by hand. My parallel and accelerated work has been CPU-side on ARM and in camera pipelines, not on GPUs. I have shipped native C++ and profiled it against hard latency limits, and I expect to pick up Rust and your DSLs quickly.

I am a Canadian permanent resident based in Toronto, ON, eligible to work without sponsorship, and available immediately. I would welcome the opportunity to discuss how my experience in real-time systems can support the High-Performance Onboard Algorithms team at Waabi. My portfolio, publications and open datasets are available at nishal.xyz.

Sincerely,

**Nishal Stanislaus Silva**, PhD